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[GS PAPER II: PUBLIC HEALTH, HEALTHCARE ACCESSIBILITY AND VULNERABLE SECTIONS. • GS PAPER III: BIODIVERSITY, HUMAN–WILDLIFE INTERACTIONS AND SCIENCE AND TECHNOLOGY.]

Publication Date: 15/10/2026

Snakebite in India: Common Krait, Rural Vulnerability and Public Health



Common krait bites highlight how sleeping conditions, rural exposure and delayed treatment contribute to India's snakebite burden. Community awareness, safer homes and timely access to appropriate healthcare are essential to prevent deaths and disabilities.

Snakebite in India: Common Krait, Rural Vulnerability and Public Health

UPSC syllabus relevance

GS Paper II: Public health, healthcare accessibility and vulnerable sections.

GS Paper III: Biodiversity, human–wildlife interactions and science and technology.

Why Snakebite Is a Public Health Concern

Snakebite is both a medical emergency and a development challenge. Its consequences depend on the snake species, venom injected, circumstances of exposure and access to appropriate treatment.

Snakebite envenoming occurs when venom injected through a bite produces harmful effects in the body. Not every snakebite causes envenoming: non-venomous snakes also bite, and venomous snakes may deliver a **"dry bite"**, without injecting venom.

The World Health Organization recognises snakebite envenoming as a **Neglected Tropical Disease (NTD)**. It disproportionately affects underserved communities where prevention, transport and treatment facilities may be inadequate.

Common Krait: A Significant Nocturnal Threat

The **common krait (*Bungarus caeruleus*)** is a highly venomous snake found across much of the Indian subcontinent. It belongs to the **Elapidae** family, which also includes cobras.

Its public health significance arises from three characteristics:

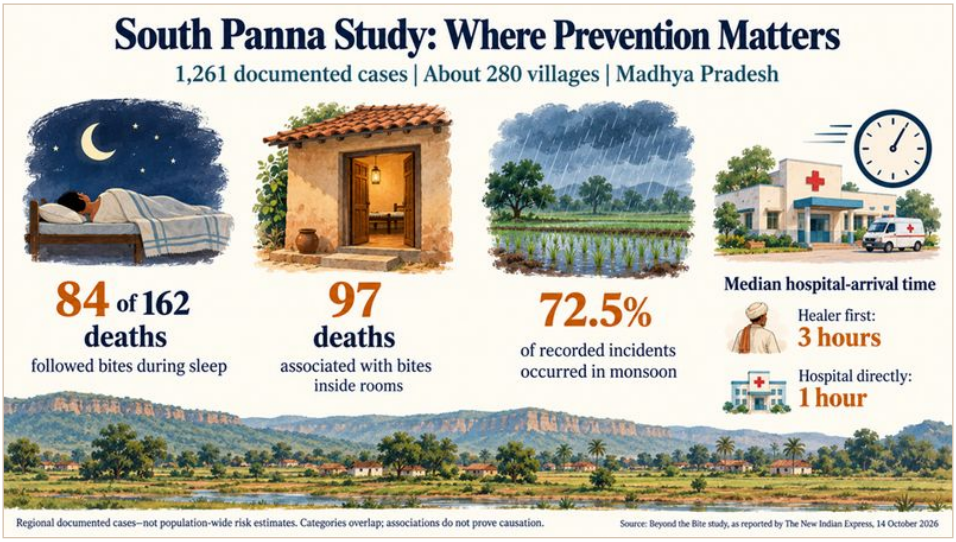
- **Nocturnal activity:** It is primarily active at night and may enter homes.
- **Limited local symptoms:** A bite may cause little immediate pain or swelling and may not wake a sleeping person.
- **Neurotoxic effects:** Its venom can disrupt communication between nerves and muscles, causing paralysis, including paralysis of breathing muscles.

Therefore, **the absence of pain or prominent bite marks does not establish that a suspected bite is harmless**.



South Panna Study: Key Findings

Median hospital-arrival time was **three hours** among those who first visited a healer, compared with **one hour** among those seeking hospital care directly. The median is the middle observation when values are arranged in order.



Interpretation: These are findings from documented cases in a particular region. They should not be treated as population-wide fatality estimates. Indoor and sleep-related categories may overlap, while observed associations do not independently establish causation.

Venomous Snakes in India

India is home to several species of venomous snakes. They are important from the perspectives of biodiversity, ecosystem balance, public health and snakebite prevention. For UPSC aspirants, the most important distinction is between snakes that can inject medically significant venom and non-venomous snakes.

Top 5 venomous snakes in India

The following five are a useful UPSC study list, covering the major medically significant snakes and two other important venomous species.



1. Indian Cobra (Naja naja)

Neurotoxic venom

Recognised by its expanded hood, the Indian cobra can cause paralysis and respiratory failure. It is one of India's medically significant venomous snakes.



2. Common Krait (Bungarus caeruleus)

Neurotoxic venom

Primarily nocturnal, it may bite people sleeping indoors. The bite can be relatively painless, while the venom can cause progressive paralysis, including respiratory muscle paralysis.



3. Russell's Viper (Daboia russelii)

Haemotoxic effects

Its venom can disrupt blood clotting, cause internal bleeding and contribute to acute kidney injury. It is a major cause of serious snakebite incidents in India.



4. Saw-scaled Viper (Echis carinatus)

Haemotoxic effects

A relatively small viper, it can cause severe bleeding and clotting abnormalities. It is particularly important in snakebite studies involving dry and semi-arid regions.



5. King Cobra (Ophiophagus hannah)

Neurotoxic venom

The world's longest venomous snake, it is primarily associated with forest habitats. Its venom can cause paralysis and respiratory failure. It mainly feeds on other snakes.

The first four species constitute India's traditionally recognised “Big Four”, which account for a major share of severe snakebite cases. The king cobra is included here as another important venomous species, not as a member of the Big Four.

Snake	Scientific name
Banded krait	Bungarus fasciatus
Monocled cobra	Naja kaouthia
Hump-nosed pit viper	Hypnale hypnale
Malabar pit viper	Craspedocephalus malabaricus complex
Bamboo pit viper	Craspedocephalus gramineus
Himalayan pit viper	Gloydius himalayanus
Sind krait	Bungarus sindanus
Sea snakes	Various species, including Hydrophis species
Andaman cobra	Naja sagittifera
Red-necked keelback	Rhabdophis subminiatus

Major venomous snakes around the world

There is no single universal ranking of the world's “top five” most dangerous snakes. Medical importance varies by region, frequency of human encounters, venom effects and access to treatment. The following are important examples:



Inland taipan — Oxyuranus microlepidotus

Australia

Known for exceptionally potent venom. Human encounters are uncommon.



Black mamba — Dendroaspis polylepis

Sub-Saharan Africa

A fast-moving snake whose neurotoxic venom can cause severe paralysis.



Saw-scaled viper — Echis species

Africa, Middle East and Asia

Medically important because of its association with serious snakebite cases in several regions.



Coastal taipan — Oxyuranus scutellatus

Australia and New Guinea

Possesses potent venom that can affect the nervous system and blood clotting.



Fer-de-lance / lancehead — Bothrops species

Central and South America

Pit vipers responsible for many medically significant bites in parts of Latin America.

Other globally important groups include rattlesnakes (Crotalus), coral snakes (Micrurus), cobras (Naja), mambas (Dendroaspis), puff adders (Bitis) and sea snakes.

How Snake Venom Affects the Body

Snake venom contains multiple biologically active substances. Its effects vary between species and may overlap.

Effect	What happens	Possible consequence
Neurotoxicity	Nerve–muscle communication is disrupted	Drooping eyelids, difficulty swallowing and respiratory paralysis
Coagulation disturbances	Normal blood clotting is disrupted	Abnormal bleeding and circulatory complications
Local tissue injury	Cells and tissues around the bite are damaged	Swelling, blistering and tissue death
Muscle injury	Muscle tissue breaks down	Weakness and possible kidney complications
Kidney injury and shock	Organ damage or reduced blood circulation develops	Acute kidney injury and organ failure

A snake's venom cannot always be classified into a single category. Describing a species as predominantly neurotoxic or haemotoxic is a useful simplification, rather than an exclusive classification.



Medically Important Venomous Snakes in India

The Traditionally Recognised "Big Four"

Snake	Scientific name	Important clinical effects
Indian cobra	*Naja naja*	Neurotoxicity; local tissue injury may occur
Common krait	*Bungarus caeruleus*	Severe neurotoxicity, sometimes with minimal local symptoms
Russell's viper	*Daboia russelii*	Clotting disturbances, bleeding and kidney injury
Saw-scaled viper	*Echis carinatus*	Clotting disturbances and bleeding

Other medically important groups include **additional krait and cobra species, pit vipers and sea snakes**.

The **king cobra** is another important venomous snake, but it is **not a member of the Big Four**. A universal ranking of the "most dangerous" snakes is misleading because medical importance depends on human exposure, venom effects and treatment access.

Why Rural Communities Face Greater Risk

Household and Occupational Exposure

Agricultural work, walking in poorly lit areas and handling material concealing snakes can increase exposure. Indoors, floor-level sleeping, gaps in buildings and surroundings that attract prey may create opportunities for encounters.

The South Panna study reported more indoor and household-associated bites among women, while men were more commonly bitten in fields or while walking. These patterns reflect local activities and exposure.

Seasonal Conditions

Monsoon conditions may change snake movement and human activity. Agricultural work and waterlogging can increase encounters in some locations, although patterns vary geographically.

Delayed Recognition and Treatment

Painless nocturnal bites may go unnoticed. Distance, transport difficulties, treatment costs and initial reliance on traditional remedies can further delay hospital care.

Traditional remedies cannot reliably neutralise venom or replace appropriate medical treatment.

Treatment: Antivenom and Supportive Care

Antivenom contains antibodies that neutralise susceptible venom toxins. Its effectiveness depends on the venom involved and the product used; an antivenom effective against one species may not adequately treat another.

Treatment may also require:

- **Respiratory support**, including mechanical ventilation for paralysis.
- Management of **bleeding, shock and kidney injury**.
- Wound care, rehabilitation and follow-up.

Antivenom is administered by trained medical professionals when clinically indicated. Its availability alone is insufficient without staff, emergency equipment and referral arrangements.



Prevention and Governance Priorities

Priority	Required measures
Safer homes and sleeping arrangements	Raised sleeping surfaces where feasible, properly tucked mosquito nets and reduced entry points
Safer surroundings	Improved grain storage, waste management and reduced rodent attraction
Occupational precautions	Torches, protective footwear and caution when handling concealed material
Community awareness	Locally understandable information on prevention and immediate referral
Accessible emergency care	Ambulances, referral networks, trained staff, appropriate antivenom and respiratory support
Better surveillance	Record deaths, disabilities, exposure circumstances and treatment delays
Rehabilitation	Support survivors facing physical disability, psychological distress or income loss

Traditional healers and community leaders can support prevention by encouraging **immediate referral to medical facilities**.

India's Policy Response

India launched the **National Action Plan for Prevention and Control of Snakebite Envenoming (NAPSE)** in **March 2024**. It provides a framework for prevention, stronger healthcare services, surveillance and community participation, with the goal of halving snakebite deaths and disability by **2030**.